

The positive exchange quotient: what completion does not forget

Draft (2026-08-10). Research report for the exchange-comparison track. This document makes no machine decision and does not edit the coefficient discipline of AIP-5.

Claim labels used throughout:

- **Verified** means checked against an accepted AIP, a mechanized repository fact, or the exact finite computation in `cocycle-classification.md` §2.4.
- **Desk-proved** means a complete mathematical argument is given here but is not mechanized.
- **Conjectured** means a precise statement whose proof is not supplied.
- **Wall** means that the requested conclusion is false as stated or stops at a named missing lemma.

1. Verdict first

Fix a finite reachable action graph $G = (V, E)$ with finite instruction alphabet Σ and edge labeling $\ell : E \rightarrow \Sigma$. Let

$$L : \mathbb{Z}^\Sigma \rightarrow \mathbb{Z}^E, \quad (L\lambda)(e) = \lambda_{\ell(e)},$$

and let $\delta\phi(e) = \phi(t(e)) - \phi(s(e))$. The formal classification group of `cocycle-classification.md` is

$$\mathcal{Q}_G = \mathbb{Z}^E / (\delta\mathbb{Z}^V + L\mathbb{Z}^\Sigma).$$

Desk-proved, main comparison theorem. The natural exchange quotient defined below maps injectively to $\mathcal{Q}_G$. Two nonnegative costs have the same formal class if and only if they are related by the natural monoid congruence. Consequently the quotient is cancellative, and there is no pair of $\mathbb{N}$ -costs that becomes equal only after group completion.

There is a stronger finite-grade conclusion. Every integral edge cost can be made nonnegative by adding a sufficiently large nonnegative price separately to each instruction. Those prices are zero in the quotient. Therefore

$$\mathcal{M}_G \cong \mathcal{Q}_G;$$

the positive edge-cost quotient is already an abelian group. For local tables, the corresponding quotient is the subgroup

$$\mathcal{M}_G^{\text{loc}} \cong q\rho(\mathbb{Z}^D) \subseteq \mathcal{Q}_G,$$

and it too is already a group. Its group completion changes nothing.

Wall, proposed “positive image” distinction. Under the stated word-price quotient, the affine semigroup $q\rho(\mathbb{N}^D)$ from `cocycle-classification.md` (11) is not an extra cone: it equals $q\rho(\mathbb{Z}^D)$. Thus group completion does **not** forget positivity here. The useful positive datum is instead the relation

$$R_c = \{(k, \lambda) \in \mathbb{N} \times \mathbb{N}^\Sigma : kc \text{ exchanges with } L\lambda\},$$

which records which formal torsion relations admit nonnegative word prices. The existing four-example computations of R_c remain substantive and are collected in §4.

Visit parity illustrates the separation exactly. Before killing word prices, twice the surcharge exchanges with the natural price “one per down.” In $\mathcal{M}_G$ this says $x \neq 0$ but $2x = 0$; in $\mathcal{Q}_G$ it says that the image of x has exact order two. Injectivity of (2) makes these the same equality viewed in a monoid presentation and in a group, not information lost by completion.

2. The natural quotient

2.1 Edge costs and local tables

The ambient version begins with the commutative monoid

$$S_G = \mathbb{N}^E.$$

For a fixed observer and grade, let $D \subseteq \Sigma \times Q$ be the reachable generator/observation pairs and let

$$\rho : \mathbb{N}^D \rightarrow \mathbb{N}^E$$

be the table-to-edge map of `local-classes.md` §5.1. The semantic local version begins with $S_G^{\text{loc}} = \rho(\mathbb{N}^D)$. A constant table $j\lambda(a, q) = \lambda_a$ satisfies $\rho(j\lambda) = L\lambda$, so both starting monoids contain every natural word-price cost.

The edge version makes the comparison with (1) transparent. Every result below restricts to the local image, and (3) states the only change.

2.2 Exact congruence

For $c, d \in S_G$, define

$$\begin{aligned} c \equiv_+ d \quad \Leftrightarrow \quad & \exists \alpha, \beta \in \mathbb{N}^\Sigma, \exists \Phi \in \mathbb{N}^V, \forall e \in E, \\ & c(e) + (L\alpha)(e) + \Phi(s(e)) = d(e) + (L\beta)(e) + \Phi(t(e)). \end{aligned}$$

There are two price vectors because the congruence permits adding a natural word price on either side. Equation (5) is entirely subtraction-free. With $\alpha = \beta = 0$ it is exactly exchange with an $\mathbb{N}$ -potential; with $\Phi = 0$ it identifies a cost with any natural word-price shift.

Desk-proved. Relation (5) is the least monoid congruence containing

1. every exchange pair $c + \Phi \circ s = d + \Phi \circ t$ with $\Phi : V \rightarrow \mathbb{N}$; and
2. every word-price pair $c \sim c + L\lambda$ with $\lambda \in \mathbb{N}^\Sigma$.

Proof. Reflexivity uses zero data. Addition compatibility follows by adding the same cost to both sides. Transitivity adds the two price vectors and the two potentials. For symmetry, choose $K \geq \max_v \Phi(v)$ and put $\Psi = K - \Phi$. Rearranging (5) without leaving $\mathbb{N}$ gives

$$d + L\beta + \Psi \circ s = c + L\alpha + \Psi \circ t.$$

Thus (5) is a monoid congruence containing the generators. Conversely, its displayed witness factors through one price shift on each side and one exchange, so every pair in (5) belongs to the generated congruence. $\square$

Define

$$\mathcal{M}_G = S_G / \equiv_+, \quad \mathcal{M}_G^{\text{loc}} = S_G^{\text{loc}} / \equiv_+.$$

This definition avoids an ambiguity in “modulo exchange and prices”: it is a congruence compatible with addition, not merely the transitive closure of a directed rewriting rule.

3. The comparison and kernel theorems

Let $q : \mathbb{Z}^E \rightarrow \mathcal{Q}_G$ be the quotient map. Sending the class of c to $q(c)$ defines

$$\iota_G : \mathcal{M}_G \rightarrow \mathcal{Q}_G.$$

It is well-defined because (5) implies

$$c - d = L(\beta - \alpha) + \delta\Phi.$$

3.1 Equality reflection

Theorem, desk-proved. For $c, d \in \mathbb{N}^E$,

$$q(c) = q(d) \iff c \equiv_+ d.$$

The same equivalence holds for two costs in the reachable local image.

Proof. The reverse implication is (8). For the forward implication, equality in (1) gives an integer price $z \in \mathbb{Z}^\Sigma$ and integer potential $\phi \in \mathbb{Z}^V$ such that

$$c - d = Lz + \delta\phi.$$

Write $z = z^+ - z^-$ coordinatewise, with $z^+, z^- \in \mathbb{N}^\Sigma$. Since V is finite, choose one constant K such that $\Phi = \phi + K$ is nonnegative on V . Constants have zero coboundary, and the last display becomes

$$c + Lz^- + \Phi \circ s = d + Lz^+ + \Phi \circ t,$$

which is the natural witness (5). No edge cost is subtracted. $\square$

Corollary, desk-proved. The map ι_G is injective. Both $\mathcal{M}_G$ and $\mathcal{M}_G^{\text{loc}}$ are cancellative. In particular,

$$a + x = b + x \Rightarrow a = b.$$

This answers the kernel question: neither direction admits a counterexample. The finiteness of V is used precisely to naturalize the integer potential. It is the fixed-grade content of AIP-5 §1C; no uniform bound on the shift K across the grade tower is asserted.

3.2 The quotient is already a group

Injectivity alone does not usually make a positive monoid a group. Killing all word prices does so here.

Theorem, desk-proved. Map (7) is an isomorphism. The local map is an isomorphism onto $q\rho(\mathbb{Z}^D)$.

Proof for edge costs. Given $c \in \mathbb{N}^E$, choose

$$\kappa_a = \max\{c(e) : \ell(e) = a\},$$

with maximum zero if no edge has label a . Then $c^\vee = L\kappa - c$ is an honest nonnegative edge cost, and

$$c + c^\vee = L\kappa \equiv_+ 0.$$

Thus every element of $\mathcal{M}_G$ has an inverse. Its image contains $q(\mathbb{N}^E)$; every integral edge cochain becomes nonnegative after a possibly larger per-letter shift, so $q(\mathbb{N}^E) = q(\mathbb{Z}^E) = \mathcal{Q}_G$.

For a local table T , take $\kappa_a = \max\{T(a, q) : (a, q) \in D\}$ and define $T^\vee(a, q) = \kappa_a - T(a, q)$. Then $T + T^\vee = j\kappa$, giving the same inverse argument inside local tables. An arbitrary integral local table is naturalized in the same way. Hence its image is exactly $q\rho(\mathbb{Z}^D)$. $\square$

Answer: what completion forgets. At a fixed finite grade, for the quotient specified in §2, nothing: the unit $\mathcal{M}_G \rightarrow \text{Gr}(\mathcal{M}_G)$ is an isomorphism, and the induced map

$$\text{Gr}(\mathcal{M}_G) \xrightarrow{\sim} \mathcal{Q}_G$$

is an isomorphism as well. For local tables, replace the target by the local formal subgroup in (3).

What is forgotten by (1) is not recovered by calling (6) positive:

- the actual nonnegative representative T and its magnitude;
- whether an equality to a word price uses a nonnegative price vector;
- the semigroup of all such witnesses (k, λ) ;
- across a tower, whether the **same** λ works at every grade.

These live before or over the quotient. The next section isolates the part that the current reports already compute.

4. Image, cone, and the four examples

4.1 The image cone collapses

Let

$$\mathcal{H}_G = \mathbb{Z}^E / \delta\mathbb{Z}^V, \quad \mathcal{H}_G^+ = \text{image}(\mathbb{N}^E \rightarrow \mathcal{H}_G).$$

This is the ordinary ordered graph-cohomology layer before word prices are killed. Let $\pi : \mathcal{H}_G \rightarrow \mathcal{Q}_G$ be the price quotient.

Desk-proved. The pushed-forward cone is all of $\mathcal{Q}_G$:

$$\pi(\mathcal{H}_G^+) = \mathcal{Q}_G.$$

For local tables,

$$q\rho(\mathbb{N}^D) = q\rho(\mathbb{Z}^D).$$

Both statements are the naturalization construction in (10). In particular, the preorder induced on the word-price quotient is indiscrete: every element and its negative are positive. Equation (13) corrects the interpretation, though not the formula, of the “full positive image” in `cocycle-classification.md` (11).

4.2 The positive relation that remains

For a named nonnegative cost c , retain

$$R_c = \{(k, \lambda) \in \mathbb{N} \times \mathbb{N}^\Sigma \mid \exists \Phi \in \mathbb{N}^V, kc + \Phi \circ s = L\lambda + \Phi \circ t\}.$$

Equivalently, before quotienting prices, kc equals the positive price $L\lambda$ in $\mathcal{H}_G$. In cycle coordinates this is the homogeneous Diophantine system $L_{\text{cyc}}\lambda = kc_{\text{cyc}}$ from `cocycle-classification.md` (13). It is a Presburger-definable affine semigroup. Unlike (12), it distinguishes an integer torsion relation from one realizable by natural prices.

Verified, imported exact computation and witnesses. Use price-vector order

$$(down0, down1, up, read, write0, write1).$$

The four named costs have the following data. Here $x = [c]$; the “named subgroup” column concerns only this class, not the whole, much larger local group.

observer	grades	named subgroup	R_c
first touch	every grade	0	$\{(k, 0) : k \in \mathbb{N}\}$
dirty write-back	$n \geq 1$	$\mathbb{Z}x$	$\{(0, 0)\}$
visit parity	$n \geq 1$	$\mathbb{Z}/2 = \{0, x\}$	$k = 2m$ and $\lambda = (m - u, m - u, u, 0, 0, 0), \quad 0 \leq u \leq m$
warmth	$n = 1$	0	$\{(k, 0) : k \in \mathbb{N}\}$
warmth	$n \geq 2$	$\mathbb{Z}x$	$\{(0, 0)\}$

The grade-one/two orders and relations were computed exactly by SNF and integer kernels in `cocycle-classification.md` §2.4–2.5. The direct first-touch and parity identities prove their all-grade rows. The uniform dirty and warmth witnesses in that report §5.1 extend their displayed nontriviality and the absence of any nonzero (k, λ) to later grades.

Every element of each displayed named subgroup has a nonnegative table representative. For dirty and warmth this may initially look surprising. If T is a zero-one surcharge, its per-letter

complement $T^\vee = j\kappa - T$ is nonnegative and represents $-x$. Repeated sums give all integer multiples. This is precisely why representability alone is not the positive invariant wanted by AIP-5.

5. Parity: divisibility and torsion

Let c_{par} charge the transitions $1 \rightarrow 0$ of the visit-parity observer, and let Φ count marks in state zero. Let $\lambda_\downarrow$ price each down instruction by one and every other instruction by zero.

Verified, direct identity. On every runnable edge,

$$2c_{\text{par}} + \Phi \circ s = L\lambda_\downarrow + \Phi \circ t.$$

There are three exact readings.

1. In the exchange quotient **before** word prices are killed, $2[c_{\text{par}}] = [L\lambda_\downarrow]$. This is the natural divisibility statement: two copies are an honest natural static price.
2. In $\mathcal{M}_G$, word prices are the trivial class, so for $x = [c_{\text{par}}]$ equation (15) is $2x = 0$.
3. In $\mathcal{Q}_G$, $2\iota_G(x) = 0$ is an ordinary group-torsion equation.

Verified, exactness of the order. The closed word

$$(\text{down0up})^2$$

returns every state to its start and has parity cost one. A natural or integer word price gives it an even value, while a potential cancels. Hence $x \neq 0$ in both quotients. Its order is exactly two.

It is safe to say “order-two torsion” in $\mathcal{Q}_G$. In $\mathcal{M}_G$ one can say more explicitly that x is a nonzero unit with $x + x = 0$; since §3 proves that $\mathcal{M}_G$ is a group, “torsion” is also literally correct there. The natural content not visible in the bare group equation is the witness $\lambda_\downarrow \geq 0$, recorded by R_c .

6. Boyle–Handelman: captured data and the extra structure

For a two-sided topological Markov shift (X, σ) , Boyle and Handelman’s ordered first cohomology is the quotient

$$H(X, \sigma) = C(X, \mathbb{Z}) / \{\xi - \xi \circ \sigma : \xi \in C(X, \mathbb{Z})\},$$

equipped with the positive cone $H(X, \sigma)^+$ consisting of classes of nonnegative continuous integer functions. This concrete definition and its attribution are restated in Matsumoto, §1, pp. 2–3. Boyle–Handelman call the result preordered in general and prove that the ordered group is a complete invariant for flow equivalence of irreducible shifts of finite type. See Boyle–Handelman (1996) and the explicit definition in Matsumoto (2015), arXiv:1501.06965.

This already captures the basic combination “integer cocycles modulo potentials, together with classes having nonnegative representatives.” For an edge shift, a one-edge locally constant function is an edge cochain and a vertex transfer function gives the graph coboundary. Thus (11) is the one-block finite-presentation slice of (16). Allowing higher-block presentations accounts for more general locally constant functions.

There are two qualifications before identifying their object with ours.

- Boyle–Handelman work with the invertible two-sided shift and, for their completeness theorem, irreducible shifts of finite type. Reachable machine graphs with reset observers can have transient edges and irreversible components. Their bi-infinite edge shifts need not retain those open-run edges. Our signed-cycle/edge-system criterion deliberately covers them.
- Their quotient removes coboundaries. It does not come with the machine’s distinguished labeling map $L : \mathbb{Z}^\Sigma \rightarrow \mathcal{H}_G$, nor does it quotient by its image.

The exact fixed-grade relationship is therefore

$$(\mathcal{H}_G, \mathcal{H}_G^+) \xrightarrow{\text{quotient by } L\mathbb{Z}^\Sigma} \mathcal{Q}_G,$$

together with the reachable local subgroup $q\rho(\mathbb{Z}^D)$. The first pair is standard ordered-cohomology data at the graph-local level. The arrow, the specified price map L , and the local-table presentation ρ are our additional fixed-grade data.

Starting with $\mathbb{N}$ does not, by itself, add a new invariant here. The positive cone in (16) already records nonnegative representatives, and after the price quotient its image is all of $\mathcal{Q}_G$ by (12). In fact the useful positive relation can be reconstructed from ordered cohomology **plus the distinguished natural price map**:

$$R_c = \{(k, \lambda) \in \mathbb{N} \times \mathbb{N}^\Sigma : k[c] = [L\lambda] \text{ in } \mathcal{H}_G\}.$$

So the honest comparison is:

datum	ordered cohomology	adic fixed grade
integer local functions modulo potentials	yes	$\mathcal{H}_G$
cone of nonnegative representatives	yes	$\mathcal{H}_G^+$
distinguished instruction-price map	not part of the invariant	L
quotient by instruction prices	not part of the invariant	$\mathcal{Q}_G$
reachable observer-table image	not part of the invariant	ρ
natural trivialization relation	recoverable after adding L	R_c
one price vector across all grades	no	tower condition
reset-observer open-run semantics	not covered by the irreducible two-sided setting	yes

The defensible difference is not “they use $\mathbb{Z}$, we use $\mathbb{N}$.” At fixed finite grade, that difference dissolves exactly. It is the structured quotient by machine instruction prices, the observer-local

image, and the uniform graded tower problem. Conversely, Boyle–Handelman’s positive cone is richer than the cone pushed into our price quotient; their framework warns us not to discard the pre-price ordered layer if lower bounds need a nondegenerate order.

7. Lean statement ladder

The current file `lean/Adic/Cocycle.lean` provides `IsCocycle`, `Exchange`, and `exchange_append`, with costs and potentials in `Nat`. The graph and local observer types do not yet exist, so the declarations below are proposed signatures rather than compiled code. They keep integer arithmetic behind an opaque `FormalClass`; all operational statements are subtraction-free.

First package the exact congruence using the existing exchange orientation:

```
def ExchangeModuloPrices
  (f g : Word → Head n → Nat) : Prop :=
  ∃ leftPrice rightPrice : Move → Nat, ∃ potential : Head n → Nat,
  Exchange (addCost f (letterCost leftPrice))
    (addCost g (letterCost rightPrice)) potential
```

The first rung is purely natural:

```
theorem exchangeModuloPrices_equivalence :
  Equivalence (@ExchangeModuloPrices n)

theorem exchangeModuloPrices_add
  (hfg : ExchangeModuloPrices f g) :
  ExchangeModuloPrices (addCost f k) (addCost g k)
```

At graph level, define `PositiveClass G` as the quotient by this congruence and expose the formal comparison without exposing integer coefficients in the theorem statement:

```
def PositiveClass.toFormal : PositiveClass G → FormalClass G

theorem toFormal_injective :
  Function.Injective (@PositiveClass.toFormal G)

theorem positiveClass_add_right_cancel
  (a b k : PositiveClass G) (h : a + k = b + k) :
  a = b
```

The proof of `toFormal_injective` may use integers internally: split the formal price vector into positive and negative parts, then shift the finite potential by its minimum. A useful public lemma states the resulting witness only in naturals:

```
theorem sameFormal_iff_naturalWitness (c d : EdgeCost G) :
  PositiveClass.toFormal [[c]] = PositiveClass.toFormal [[d]] ↔
  ∃ α β : LetterPrice G, ∃ Φ : Vertex G → Nat,
  ∀ e, c e + α (label e) + Φ (source e) =
    d e + β (label e) + Φ (target e)
```

The result that the monoid is already a group should also be stated in exchange form:


```

theorem exists_natural_inverse (c : EdgeCost G) :
  ∃ cInv : EdgeCost G, ∃ price : LetterPrice G,
    Exchange (edgeCost (c + cInv)) (letterCost price) 0

```

```

theorem positiveClass_isAddGroup :
  ∀ x : PositiveClass G, ∃ y, x + y = 0

```

For local tables, `cInv` is the per-letter complement of the table, so this theorem requires no `Int`. The local comparison target should be named `LocalFormalClass G`, preventing the ambient/local-image confusion warned about in `cocycle-classification.md` §2.1.

Finally parity separates the positive witness from formal torsion:

```

theorem parity_twice_exchange :
  Exchange (fun word config => 2 * parityCost word config)
    (letterCost downPrice) parityPotential

```

```

theorem parity_not_exchange :
  ¬ ∃ price potential,
    Exchange parityCost (letterCost price) potential

```

```

theorem parity_positiveClass_order_two :
  parityClass ≠ 0 ∧ parityClass + parityClass = 0

```

These are the main comparison results in the vocabulary of AIP-5 §1B. The only essential finiteness hypothesis enters natural symmetry and `sameFormal_iff_naturalWitness`, where an integer potential is shifted into `Nat`.

8. Consequences and boundary

1. **Desk-proved.** “Positive cost quotient” is algebraically safe but misleading if read as a nontrivially ordered monoid: after quotienting all word prices it is a group with total image cone.
2. **Desk-proved.** SNF loses no equality between natural costs at a fixed grade. It loses the sign of a trivializing price witness unless R_c (or the pre-price ordered group with distinguished L) is retained.
3. **Verified for the examples.** Parity is the minimal arithmetic case: exact group order two and an explicit natural doubled-price witness. Dirty and warmth are infinite-order scheduling cases; first touch is zero.
4. **Wall, tower extrapolation.** The proof naturalizes a potential using a grade-dependent finite maximum. It does not provide a uniform potential bound or a single price vector across grades. The tower wall remains the separate one already named in `cocycle-classification.md`.
5. **Recommendation.** Preserve $(\mathcal{H}_G, \mathcal{H}_G^+)$ and the distinguished map L until after positive lower-bound questions are asked. Derive $\mathcal{Q}_G$ for formal classification, and derive R_c for realizable pricing. Do not describe $q\rho(\mathbb{N}^D)$ alone as the retained positive layer.

Repository anchors

- AIP-5 §§0–2: natural exchange form and the fixed-grade bounded-potential observation.

- `aips/draft/local-classes.md` §§3–5: observer tables, exchange feasibility, parity, and the original class-monoid formulation.
- `aips/draft/cocycle-classification.md` §§1–2: $\mathcal{Q}_G$, SNF, the four exact computations, and R_c .
- `aips/draft/cost-cohomology-priorart.md` §§5.3 and 12: the Boyle–Handelman warning and the coefficient-comparison action.
- `lean/Adic/Cocycle.lean`: current Nat-valued Exchange vocabulary.

References

- M. Boyle and D. Handelman, “Orbit Equivalence, Flow Equivalence and Ordered Cohomology”, *Israel Journal of Mathematics* 95, 169–210, 1996.
- K. Matsumoto, “Continuous Orbit Equivalence, Flow Equivalence of Markov Shifts and Circle Actions on Cuntz–Krieger Algebras”, *Mathematische Zeitschrift* 285, 121–141, 2017; §1 explicitly restates the ordered-cohomology group and its positive cone.